

Ex.No. 12: NB-IoT vs LTE-M Throughput Simulation Date

OBJECTIVE :

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
% SNR values (dB)
```

```
SNR = 0:5:30;
```

```
% Simulated Throughput (Mbps)
```

```
NB_IoT = 0.05 * (1 - exp(-SNR/8));
```

```
LTE_M = 1.2 * (1 - exp(-SNR/8));
```

```
% Display values
```

```
disp('SNR (dB)  NB-IoT (Mbps)  LTE-M (Mbps)')
```

```
disp([SNR' NB_IoT' LTE_M'])
```

```
figure
```

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```
plot(SNR, NB_IoT, '-o', 'LineWidth', 2)
```

```
title('NB-IoT Throughput vs SNR')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Throughput (Mbps)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

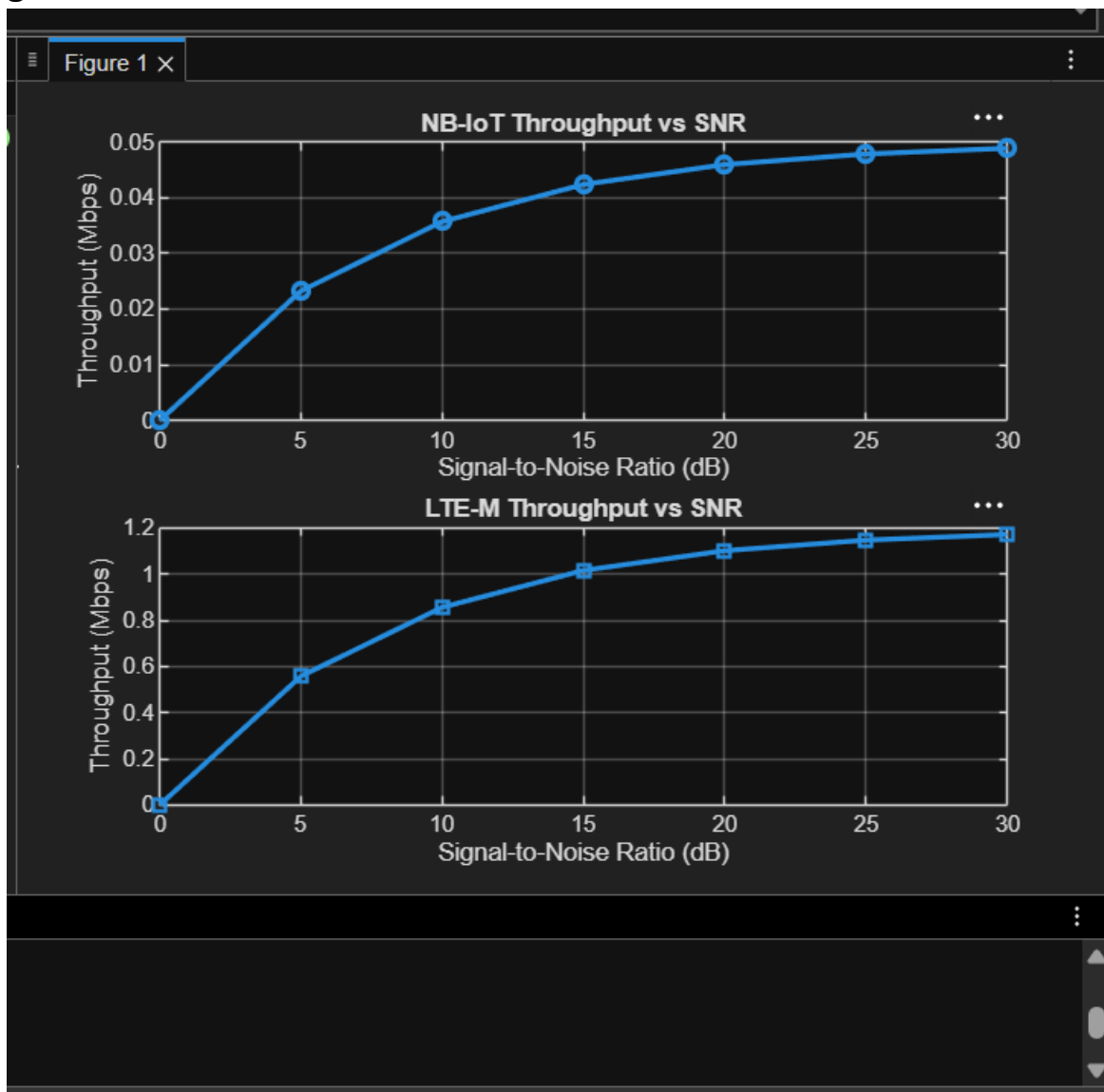
```
plot(SNR, LTE_M, '-s', 'LineWidth', 2)
```

```
title('LTE-M Throughput vs SNR')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Throughput (Mbps)')
```

grid on



OBJECTIVE 2:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

% SNR values (dB)

```
SNR = 0:5:30;
```

```

% Simulated Latency (ms)
NB_IoT = [180 160 145 130 120 110 100];
LTE_M = [90 80 70 60 50 45 40];

% Display values
disp('SNR(dB)  NB-IoT Latency(ms)  LTE-M Latency(ms)')
disp([SNR' NB_IoT' LTE_M'])

figure

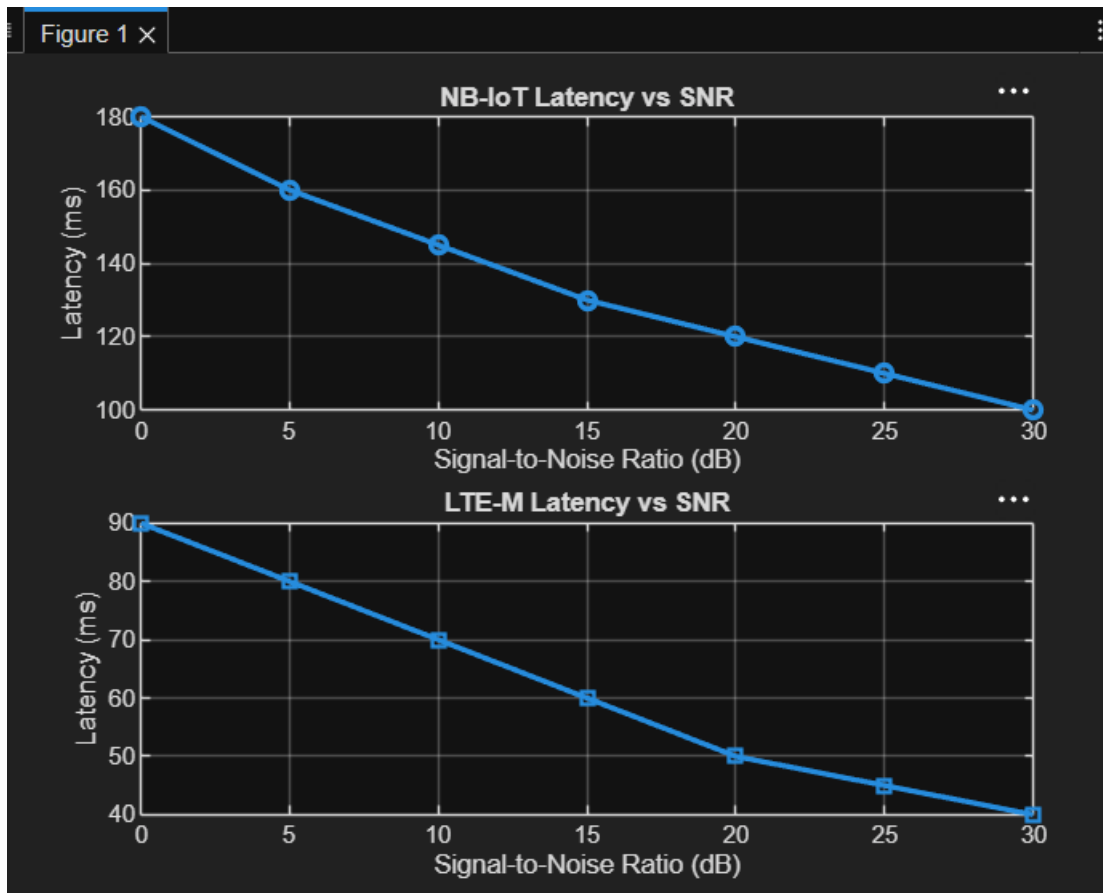
% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR, NB_IoT, '-o', 'LineWidth', 2)
title('NB-IoT Latency vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Latency (ms)')
grid on

% ----- Graph 2 -----
subplot(2,1,2)
plot(SNR, LTE_M, '-s', 'LineWidth', 2)
title('LTE-M Latency vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')

```

```
ylabel('Latency (ms)')
```

```
grid on
```



OBJECTIVE 3:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
% SNR values (dB)
```

```
SNR = 0:5:30;
```

```
% Simulated Packet Delivery Ratio (%)
```

```
NB_IoT = [82 86 89 92 95 97 99];
```

```
LTE_M = [88 91 94 96 98 99 100];  
disp('SNR(dB) NB-IoT PDR(%) LTE-M PDR(%)')  
disp([SNR' NB_IoT' LTE_M'])
```

```
figure
```

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```
plot(SNR,NB_IoT,'-o','LineWidth',2)
```

```
title('NB-IoT Packet Delivery Ratio vs SNR')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Packet Delivery Ratio (%)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

```
plot(SNR,LTE_M,'-s','LineWidth',2)
```

```
title('LTE-M Packet Delivery Ratio vs SNR')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Packet Delivery Ratio (%)')
```

```
grid on
```

Figure 1 ×

